



Gaming on your Wrist

Gameband smartwatch lets you play simple games on a wrist-mounted screen.
<http://dgit.in/Gameband1>



Rico is Aquaman!

Justice League's Aquaman actor, Jason Momoa to star in 'Just Cause' movie.
<http://dgit.in/JustCauseM>

You need exactly this much bandwidth

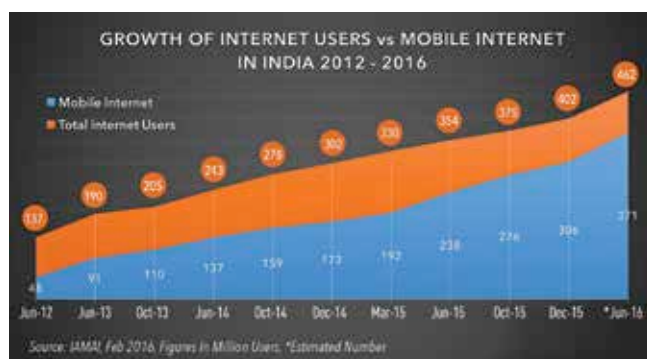
Back in the day, a 56K dialup connection was enough to take of your IM needs and to navigate many text-heavy websites. But you have 2017's rich web content to contend with today: A 1 Mbps connection loads most modern websites at a tolerably fast rate. It's enough to stream MP3-quality audio. It's also the bare minimum to stream SD-quality videos. Go up a notch to 2 Mbps and you're able to stream 720p videos. Considering the sizes of games these days, this is also the bare minimum if you expect to download anything off Steam before the end of the century. At 4 Mbps—the apparent national average—you can just about manage

from mobile devices. The main Indian use case, then, is mobile access to social media. If we take that into consideration, the connection speed question (and the reliability and latency questions) become a lot less clear: we're not talking about users on 4 Mbps broadband connections here. We're talking about people on 2G, 3G, and 4G connections.

Unfortunately, despite Jio's efforts to offer 4G in exchange for your soul for kinda-sorta free, the majority of these mobile internet users are on the slower variety of connection. Fortunately, most social media use-cases, apart from live streaming and video uploads, don't require much raw bandwidth. Also, "Lite" apps and sites like Facebook Lite ensure that you get a reasonable

mobile millennial crowd, internet use means a lot more than checking Facebook and paying your gas bill. We're looking at fundamentally different use cases here, not just in the present, but in the immediate future. Do Indian internet connections have what it takes to cater to these requirements? What are they? If YouTube usage statistics are anything to go by, video streaming is one such use case. It's here that connectivity starts meaning a lot more than peak bandwidth figures. Let's frame this simply: for a video to stream, you not only need to have connection with enough bandwidth to handle the stream without buffering, you need to have a stable connection that provides the necessary bandwidth for as long as your video's being streamed.

While both mobile and broadband users might experience good peak broadband speeds, load variations mean that your streaming experience can differ significantly at different points in time. Personally, this writer remembers spending good money on a BSNL EVDO connection to avail their promised FUP-free connection at "up to 3.5 Mbps." That "up to" is important: in practice, we never experienced speeds that were significantly higher than EDGE during the day, with occasional peaks around 1 Mbps. The situation might not be quite as extreme with most broadband or mobile connections, but your actual connection speed at a given point of time is often just a fraction of peak throughput. With that in mind, a 4 Mbps connection may or may not be enough to take care of your basic streaming needs. If reliability was the only issue, we'd still be quite happy. It isn't, however



Internet adoption rates in India over time (COURTESY: DAZEINFO)

a 1080p stream. 4-5 Mbps is also the minimum for a decent experience with cloud gaming and desktop solutions like LiquidSky and GeForce Now. If your bandwidth-intensive use cases end with 1080p video streams, a 4 Mbps connection is more than enough to provide you with a good browsing experience. This is all assuming low latency and a connection that doesn't conk off at the slightest excuse.

Indian Internet Use Cases

Let's put this another way: is a 4 Mbps connection enough to ensure a decent internet experience for an average Indian's use cases? What do Indians actually use the internet for, anyways? As of last year close to 200 million Indians were on Facebook, the most from any country. The majority of these users accessed social media content

documentation, registrations, and payments online. From sad experience, the majority of these portals look like they were designed before Y2K. Although, that means you don't need much bandwidth to avail the government's entire bouquet of digital offerings. So, social media consumption and filing government documents: The way things stand, a 4 Mbps connection is more than enough for an average Indian's internet use cases, even if that's a metered connection that's less reliable than a Jio voice call. But, what about what, say, an average reader of Digit needs in terms of internet connectivity? Well, that changes things. A lot.

Speed can matter

YouTube has over 60 million unique users in India, 70 percent of whom are under the age of 35. For the upwardly

social media experience, even if you're on a 2G connection. What else do Indians use the internet for? Under the aegis of the Digital India program, both state and central governments bodies are shifting

Bandwidth Capping: The Real Problem.

Bandwidth capping is the Indian internet's worst bugbear. This, more than anything else, shapes people's usage patterns. It only takes a 2 Mbps downlink to stream a YouTube video in 720p. The flipside to this is that you use up over 100 MB to stream just ten minutes of video content. If you limit